Analisi dati per Enrico

Descriptive statistics

Data Quality Summary

Check	Finding
Experimental runs	81
Process factors	4
Response variables	5
Duplicate records	0
Missing extraction-yield results	0
Missing fatty-acids results	18 (22.2%)

The 18 missing fatty-acid measurements are **not randomly distributed**:

- All occur at **60 bar**.
- Their extraction yields range from **0.09% to 0.62%**.
- Mean yield for these runs is **0.23%**, versus **4.02%** for runs with fatty-acid measurements.

Fatty-acid composition was unavailable at 60 bar because insufficient material was extracted. These values should therefore be labelled as something like **"not measured due to insufficient extract"**, rather than automatically imputed.

Distribution of experimental factors

This experimental design uses three levels for each factor:

Table 3: Core descriptive statistics for the experimental responses

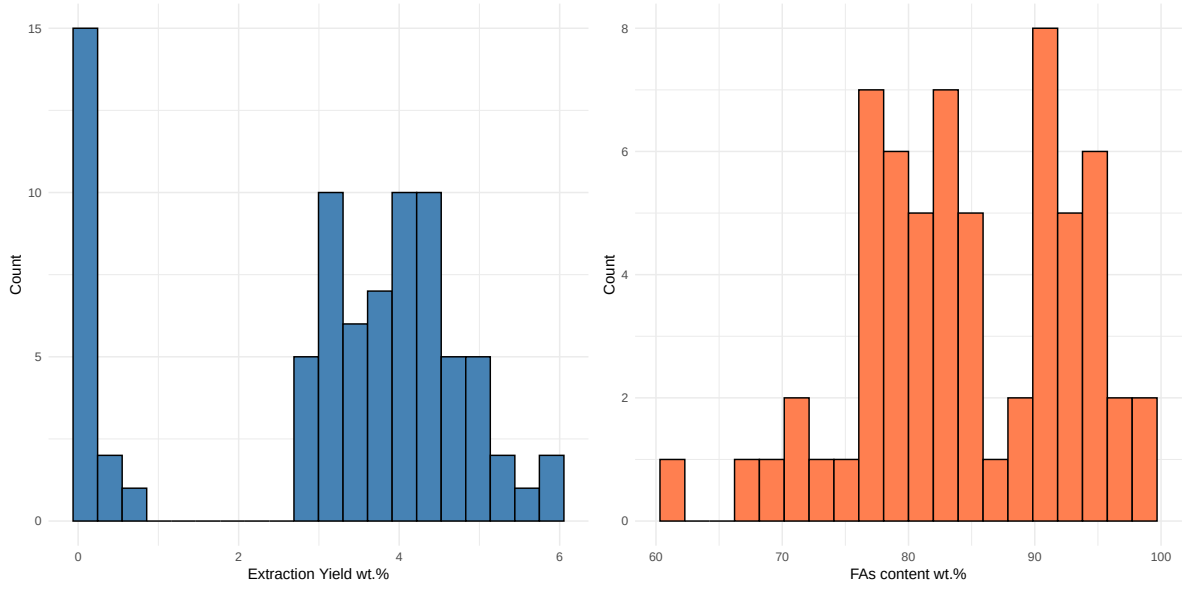
Statistic	Extraction Yield wt.%	FAs content wt.%	% C 16:0	% C 18:1	% C 18:0
Valid n	81.00	63.00	63.00	63.00	63.00
Mean	3.18	84.47	15.84	33.24	32.79
SD	1.73	8.39	1.66	3.39	3.10
Median	3.67	83.80	15.43	32.47	32.19
Q1	2.83	78.30	14.86	31.33	30.80
Q3	4.30	91.80	16.71	34.77	34.17
Minimum	0.09	61.30	12.72	25.83	26.95
Maximum	5.90	98.70	21.08	39.65	39.11

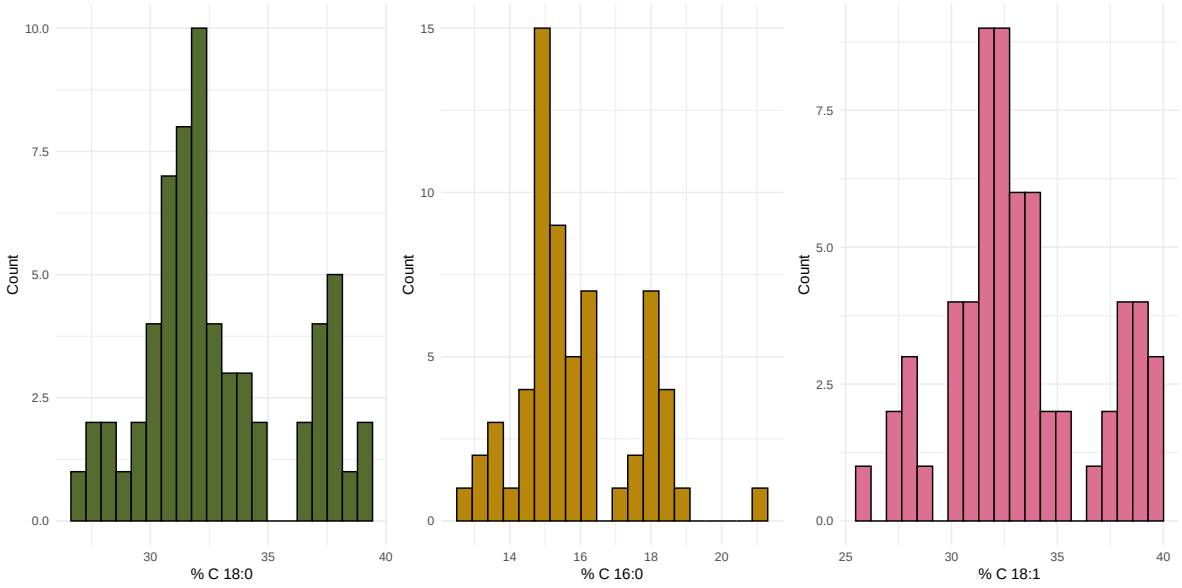
Factor	Levels	Distribution
Pressure	60, 330, 600 bar	18, 45, 18 runs
Temperature	303,333, 363 K	18, 45, 18 runs
Flow	4, 7, 10 L/min	18, 45, 18 runs
Time	30, 75, 120 mins	18, 45, 18 runs

Descriptive statistics

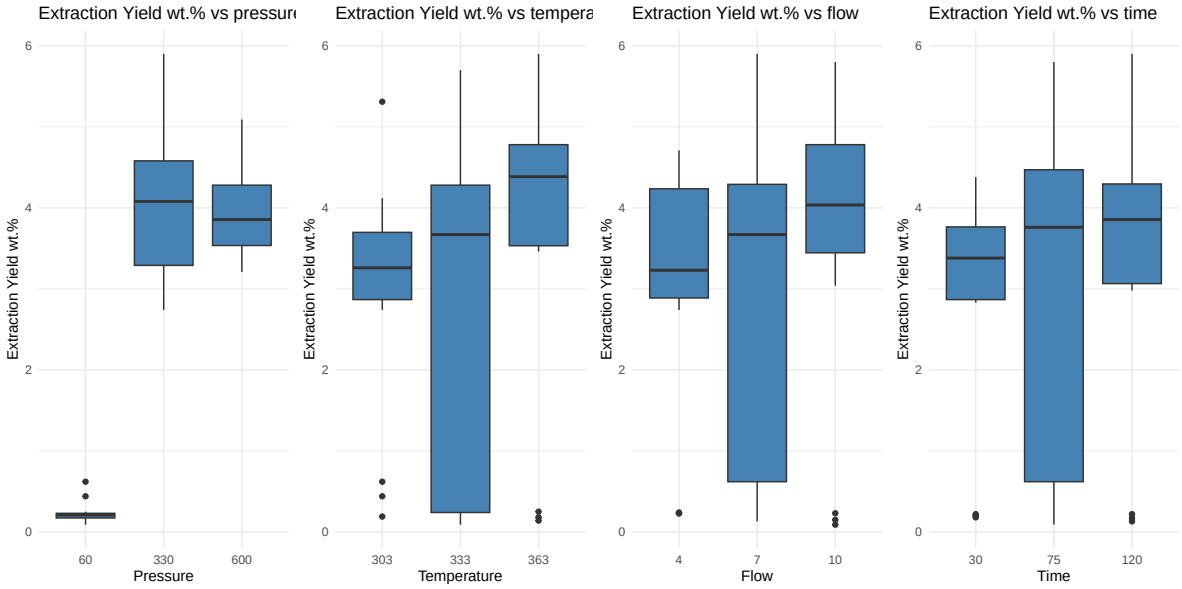
Distribution plots

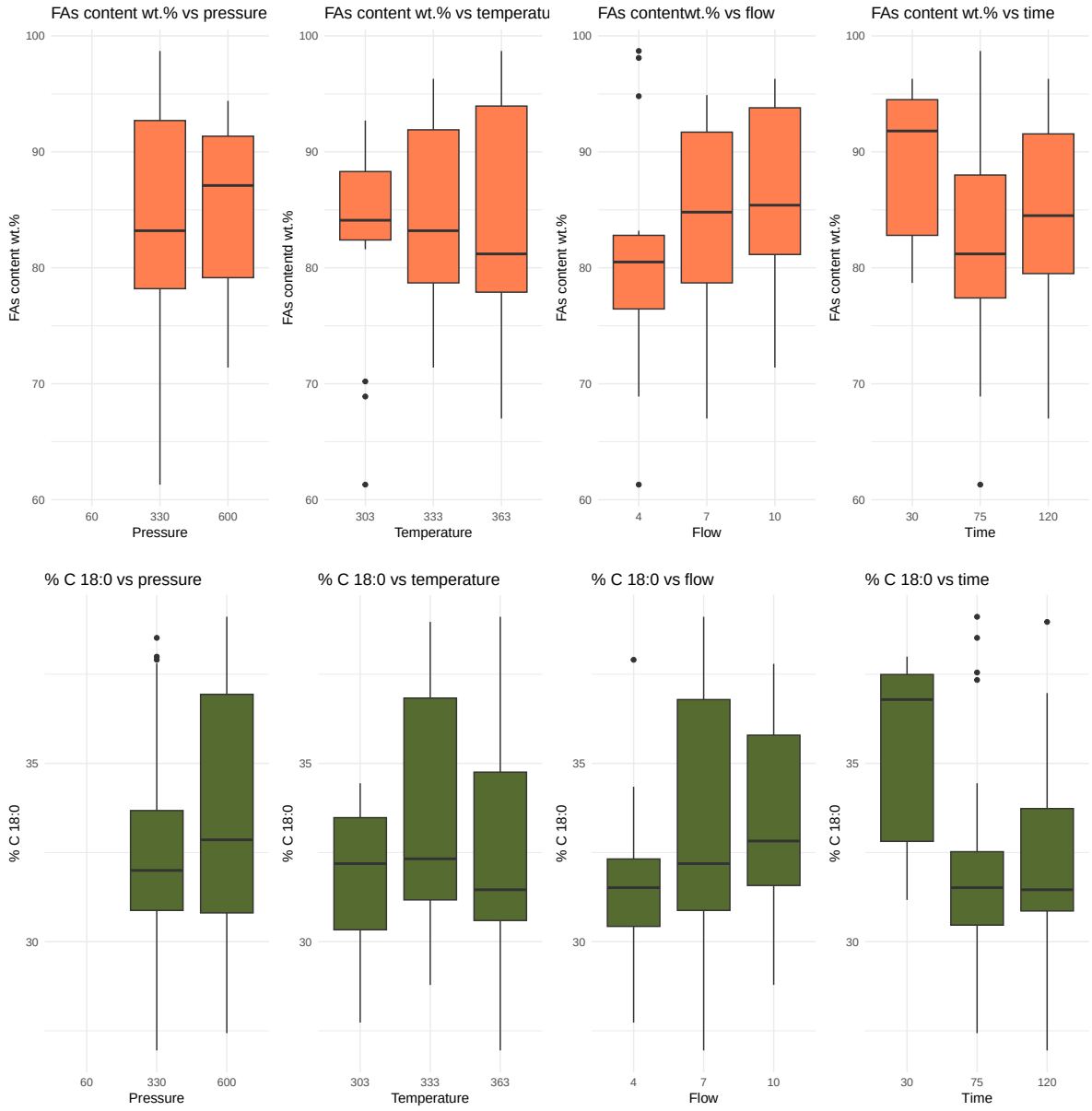
Histograms

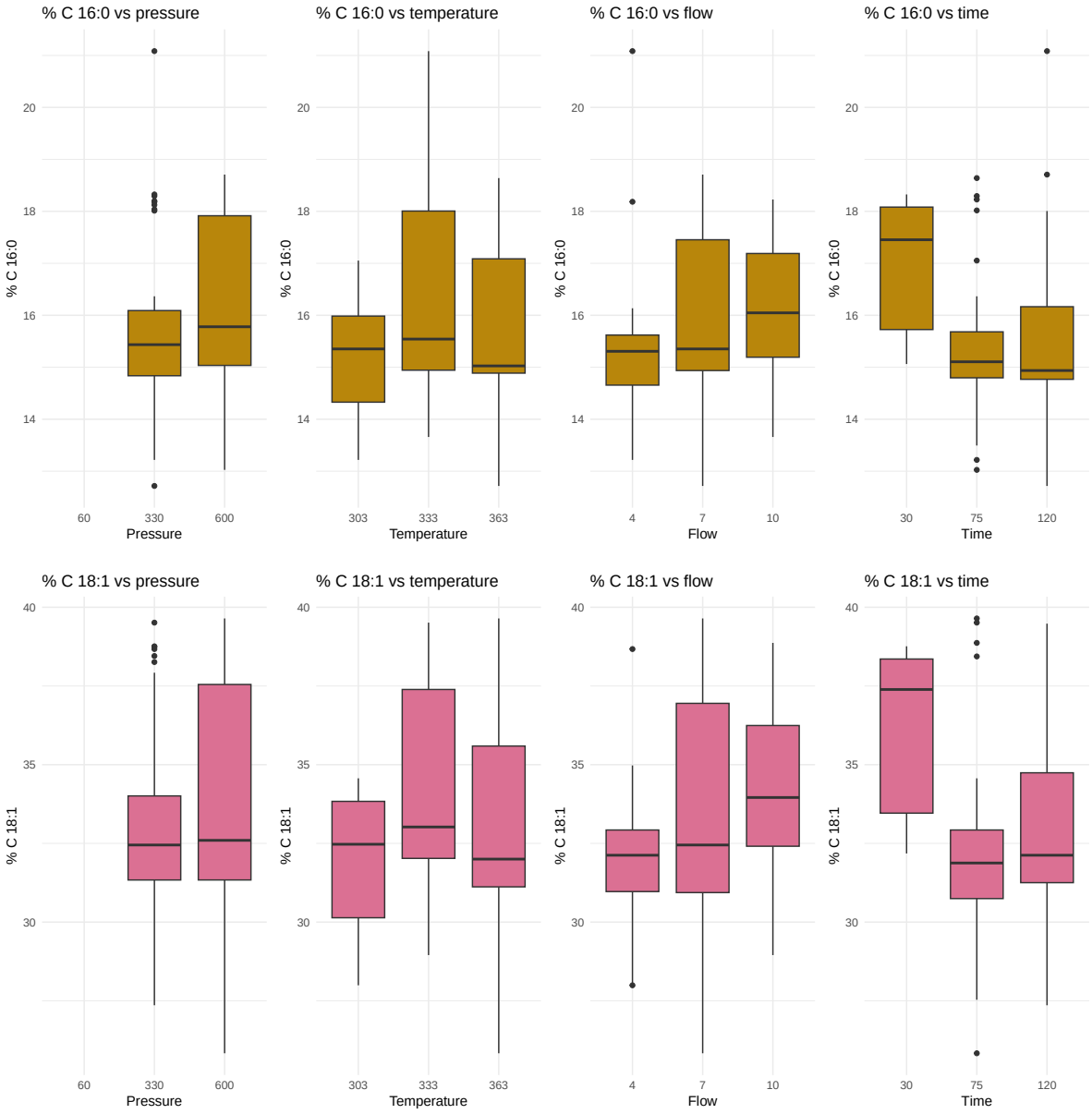



Count
% C 18:0
% C 16:0
% C 18:1

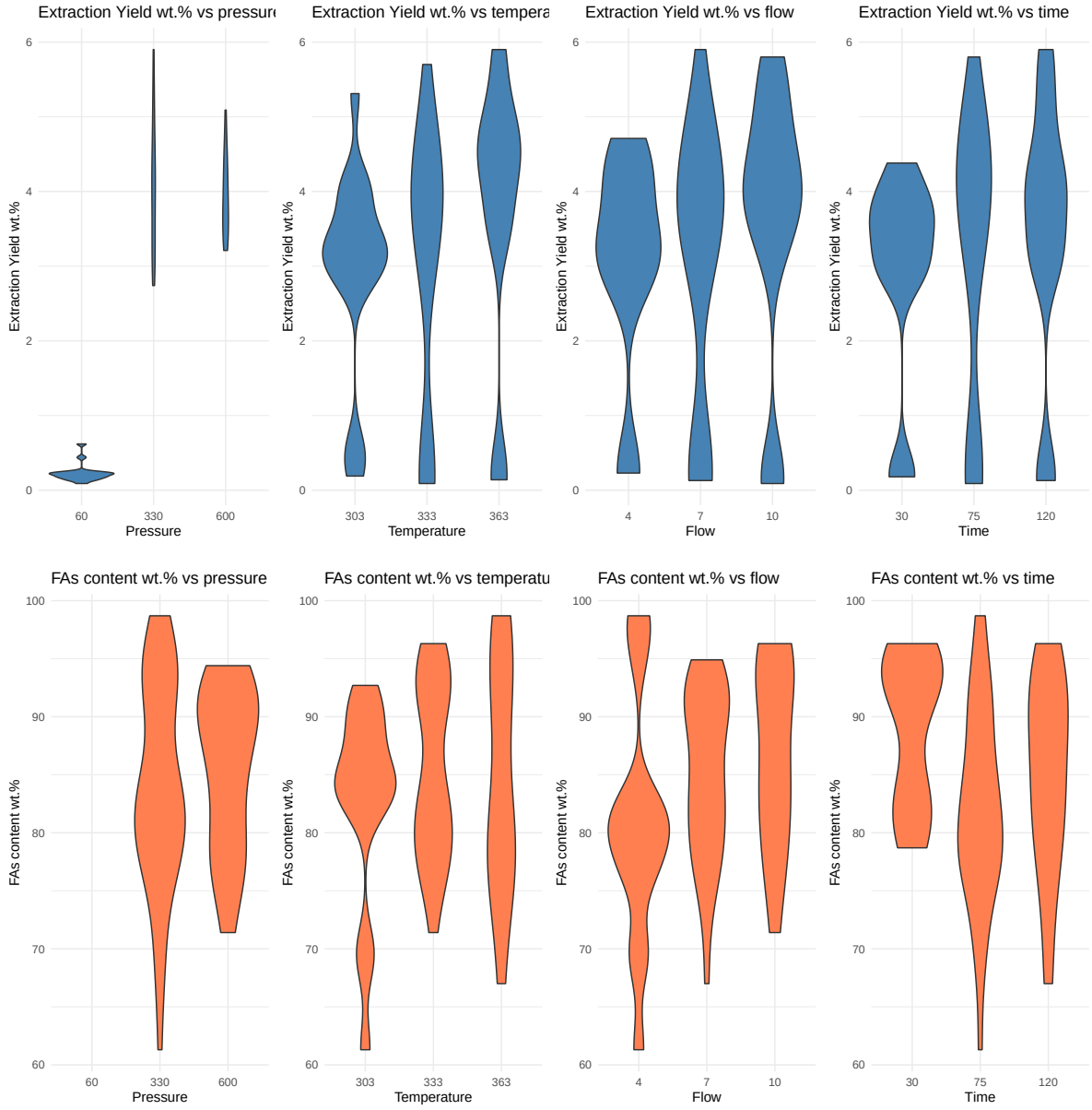
Boxplots

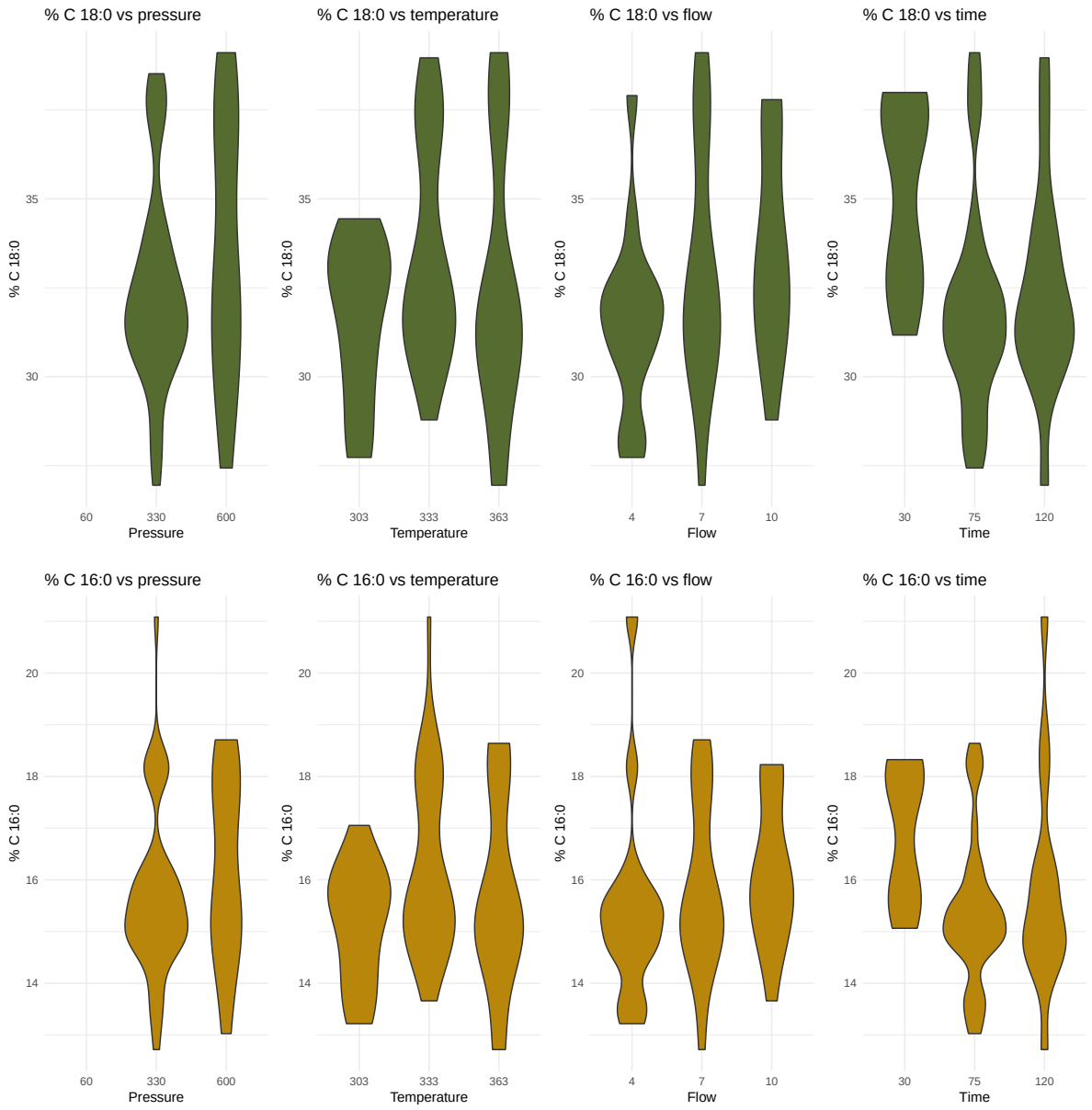

Extraction Yield wt.% vs pressure
Extraction Yield wt.% vs tempera
Extraction Yield wt.% vs flow
Extraction Yield wt.% vs time
Extraction Yield wt.%
Pressure
Temperature
Flow
Time

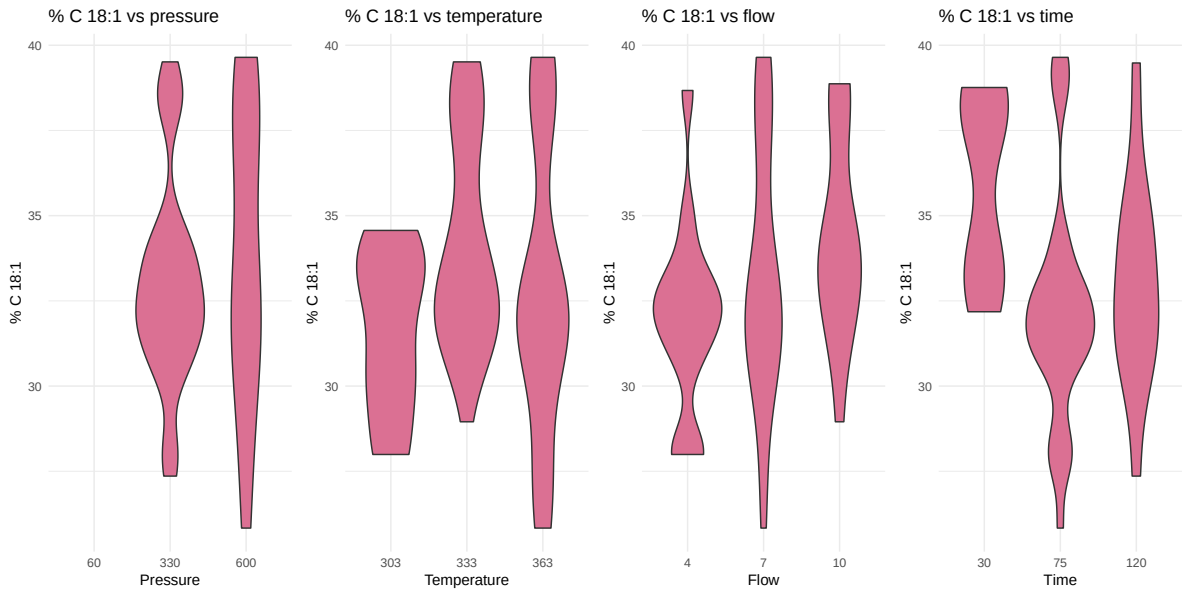

FAs content wt.% vs pressure
FAs content wt.%
100
90
80
70
60
60
330
600
Pressure
FAs content wt.% vs temperatu
FAs contentd wt.%
100
90
80
70
60
303
333
363
Temperature
FAs contentwt.% vs flow
FAs content wt.%
100
90
80
70
60
4
7
10
Flow
FAs content wt.% vs time
FAs content wt.%
100
90
80
70
60
30
75
120
Time
% C 18:0 vs pressure
% C 18:0
35
30
60
330
600
Pressure
% C 18:0 vs temperature
% C 18:0
35
30
303
333
363
Temperature
% C 18:0 vs flow
% C 18:0
35
30
4
7
10
Flow
% C 18:0 vs time
% C 18:0
35
30
30
75
120
Time


% C 16:0 vs pressure
% C 16:0 vs temperature
% C 16:0 vs flow
% C 16:0 vs time
% C 16:0
20
18
16
14
60
330
600
Pressure
303
333
363
Temperature
4
7
10
Flow
30
75
120
Time
% C 18:1 vs pressure
% C 18:1 vs temperature
% C 18:1 vs flow
% C 18:1 vs time
% C 18:1
40
35
30

Violin plots

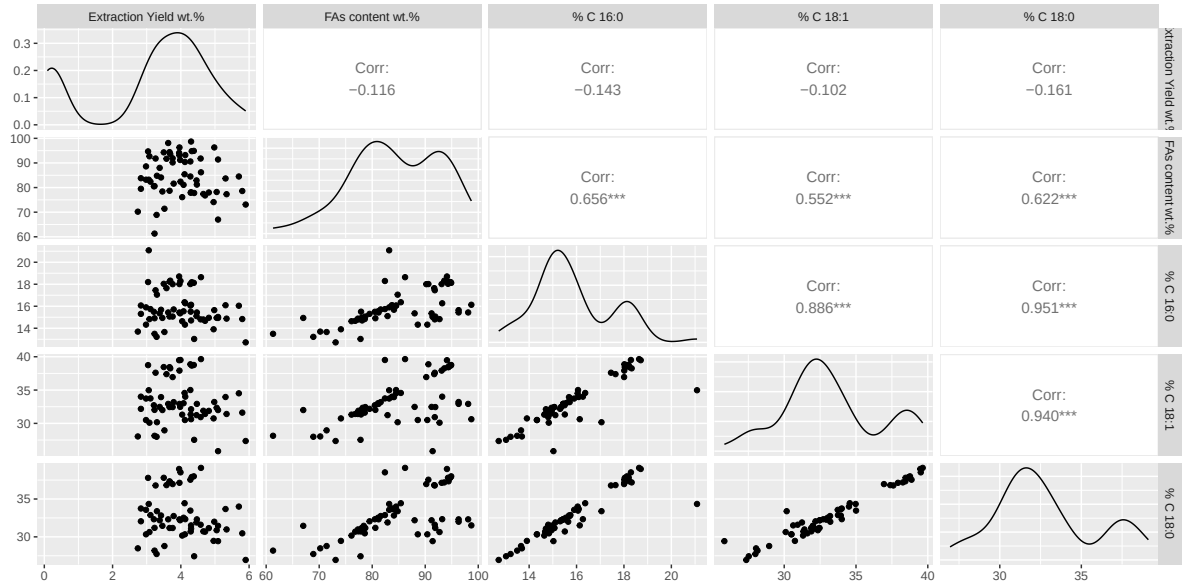



% C 18:0 vs pressure
% C 18:0 vs temperature
% C 18:0 vs flow
% C 18:0 vs time
% C 18:0
35
30
60
330
600
Pressure
303
333
363
Temperature
4
7
10
Flow
30
75
120
Time
% C 16:0 vs pressure
% C 16:0 vs temperature
% C 16:0 vs flow
% C 16:0 vs time
% C 16:0
20
18
16
14



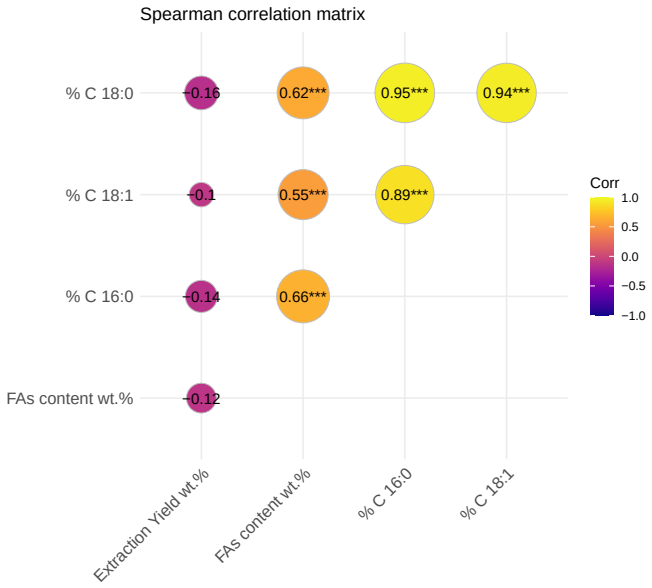
Correlation plots

```
library(GGally)
ggpairs(rawdata[, 9:13], upper = list(continuous=wrap("cor", method="spearman")))
```



Another type of correlation plot

```
library(ggcorrplot)
corr <- cor(rawdata[, 9:13], method = "spearman", use = "pairwise.complete.obs")
p.mat <- cor_pmat(rawdata[, 9:13], method = "spearman")
ggcorrplot(corr, type = "upper", p.mat = p.mat, insig='stars', method = 'circle',
           lab = TRUE, circle.scale = 2, colors = viridis::plasma(50),
           title = "Spearman correlation matrix")
```



Modelling

Model 1

Linear mixed effect model for total fraction of fatty acids, with replicate as random effect and removing observations with missing values:

```
library(nlme)
mdl1 <- nlme::lme(fixed = FA ~ Pressure + Temperature + Flow + Time,
                  random = ~ 1 | Replicate,
                  data = df,
                  method = "REML", na.action = na.omit,
                  # compound symmetry, because presumably the correlation
                  # between replicates should always be the same
                  correlation = corCompSymm(form = ~ 1 | Replicate))
summary(mdl1)
```

```
Linear mixed-effects model fit by REML
  Data: df
       AIC     BIC    logLik
  469.1144 485.598 -226.5572

Random effects:
 Formula: ~1 | Replicate
        (Intercept) Residual
StdDev:   0.4876703 8.166596

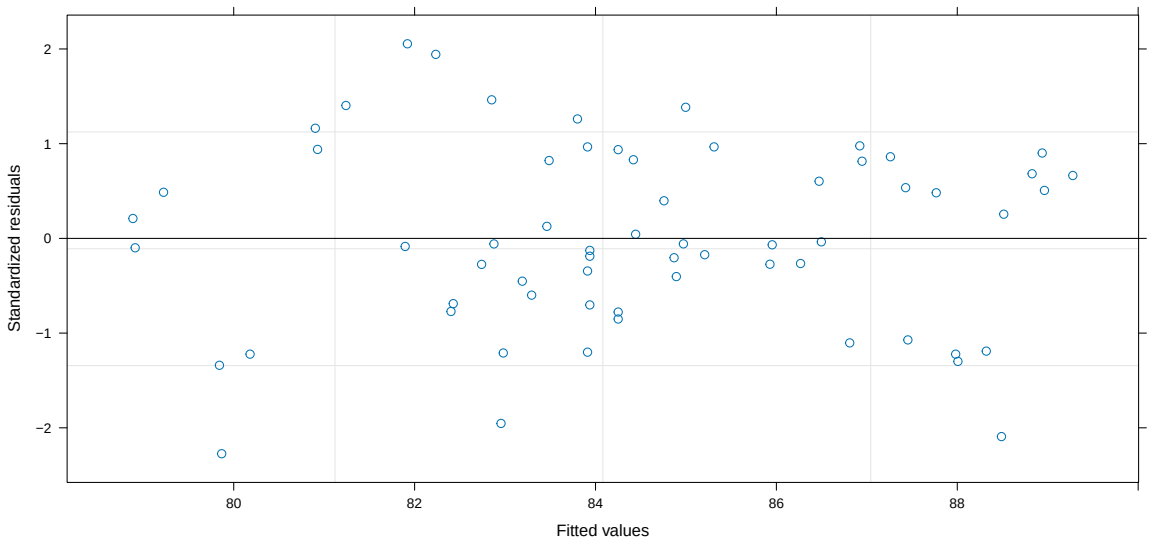
Correlation Structure: Compound symmetry
 Formula: ~1 | Replicate
 Parameter estimate(s):
        Rho
-0.02872741
Fixed effects:  FA ~ Pressure + Temperature + Flow + Time
               Value Std.Error DF   t-value p-value
(Intercept) 66.96770 17.698749 56  3.783754  0.0004
Pressure     0.00568  0.008556 56  0.663770  0.5096
Temperature  0.03422  0.050409 56  0.678889  0.5000
Flow         1.01444  0.504091 56  2.012421  0.0490
Time        -0.04407  0.033606 56 -1.311490  0.1950
 Correlation:
            (Intr) Pressr Tmprtr Flow
Pressure    -0.197
Temperature -0.948  0.000
Flow        -0.199  0.000  0.000
Time        -0.142  0.000  0.000  0.000

Standardized Within-Group Residuals:
        Min          Q1         Med          Q3         Max
-2.27354442 -0.69577563 -0.05842317  0.81831913  2.05465639

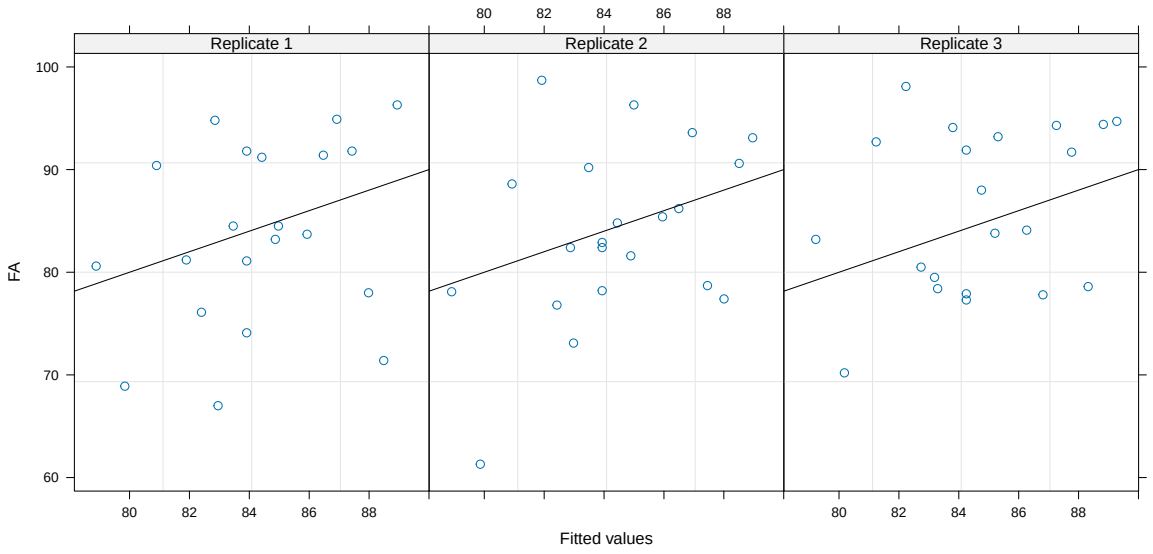
Number of Observations: 63
Number of Groups: 3
```

Various diagnostic plots for `mdl1`:

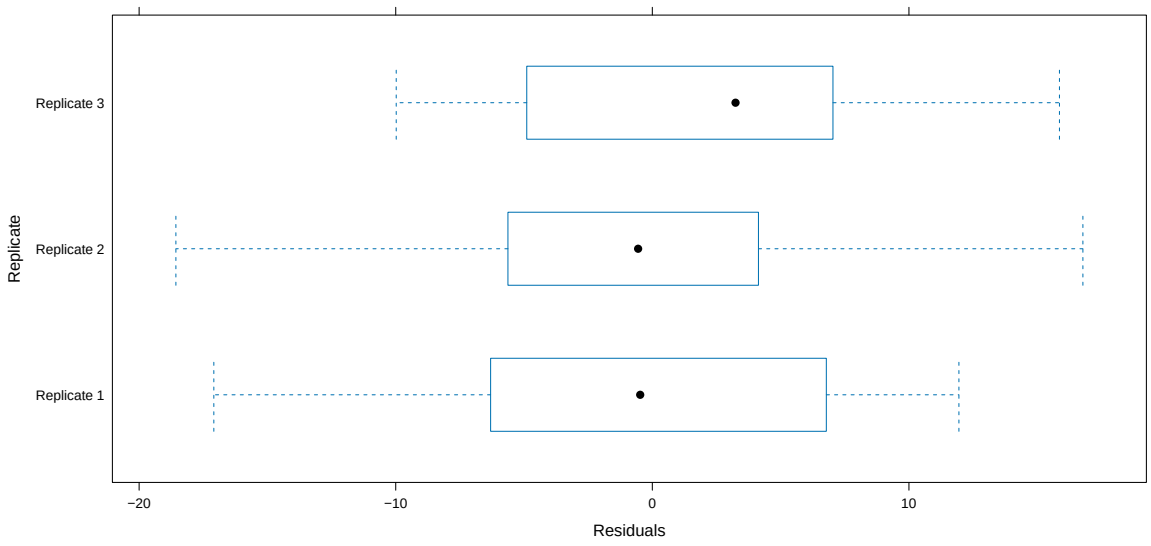
```
# residuals vs fitted (default), should be uniformly scattered
# around the zero line
plot(mdl1)
```




```
# observed vs fitted, should be distributed around the identity line
plot(mdl1, FA ~ fitted(.) | Replicate, abline = c(0,1))
```



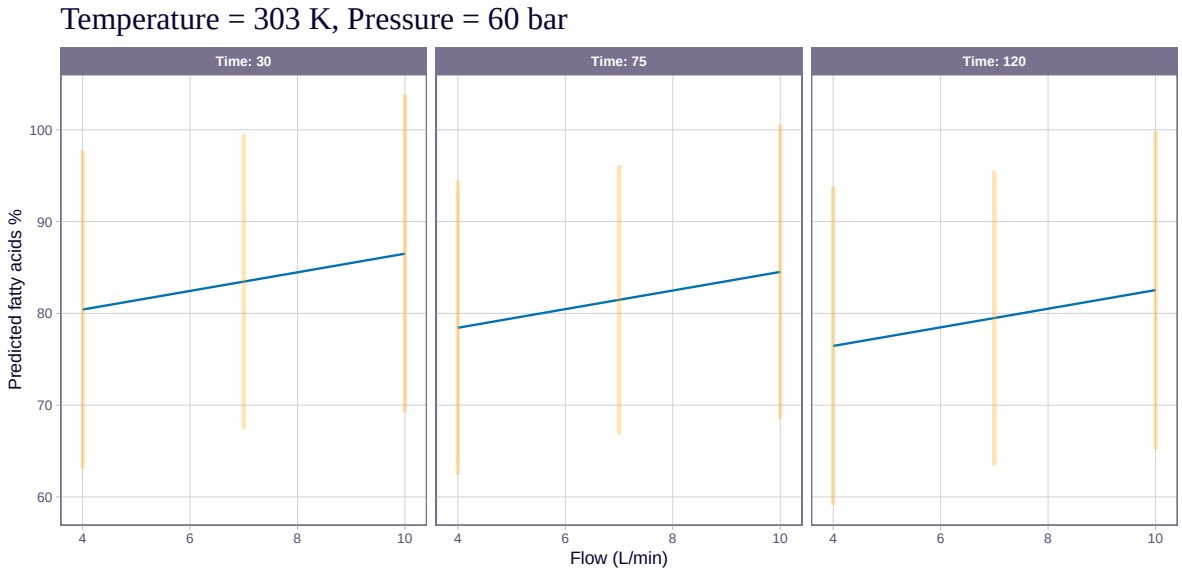

```
# boxplot of residuals by replicate, should all be centred around zero
# and with similar variance
plot(mdl1, Replicate ~ resid(.))
```



Estimated marginal means of total fatty acids percentage, at fixed temperature of 303 K and pressure of 60 bar, for flow rates of 4, 7, and 10 L/min, and times of 30, 75, and 120 minutes:

```
library(emmeans)
emmip(
  mdl1,
  ~ Flow | Time,
  CIs = T,
  at = list(
    Pressure= c(60),
    Flow = c(4, 7, 10),
    Time = c(30, 75, 120),
    Temperature = c(303)
  ),
  type = 'response',
  CIarg = list(
    linewidth = 1.5,
    alpha = 0.3,
    color = "orange"
  ),
  dotarg = list(
    size = 2,
    shape = "square",
    color = "black"
  ),
```

```
  facetlab = 'label_both',
  ylab = "Predicted fatty acids %",
  xlab = "Flow (L/min)"
) + ggplot2::labs(title = 'Temperature = 303 K, Pressure = 60 bar')
```



Numerical values of the estimated marginal means (i.e., predictions):

```
 Flow Time yvar   SE df  LCL   UCL tvar xvar
    4   30 80.4 4.03  2 63.1  97.7 1       4
    4   75 78.4 3.73  2 62.4  94.5 1       4
    4  120 76.4 4.03  2 59.1  93.8 1       4
    7   30 83.5 3.73  2 67.4  99.5 1       7
    7   75 81.5 3.41  2 66.8  96.2 1       7
    7  120 79.5 3.73  2 63.4  95.5 1       7
   10   30 86.5 4.03  2 69.2 103.8 1      10
   10   75 84.5 3.73  2 68.5 100.6 1      10
   10  120 82.5 4.03  2 65.2  99.9 1      10

Degrees-of-freedom method: containment
Confidence level used: 0.95
```

Model 2

Linear mixed effect model for fraction of C 18:0, with total fraction of FAs as covariate, replicate as random effect, and removing observations with missing values:

```
mdl2 <- nlme::lme(fixed = C180 ~ Pressure + Temperature + Flow + Time + FA,
            random = ~ 1 | Replicate,
            data = df,
            method = "REML", na.action = na.omit,
            correlation = corCompSymm(form = ~ 1 | Replicate))
summary(mdl2)
```

```
Linear mixed-effects model fit by REML
  Data: df
      AIC      BIC    logLik
  332.917 351.3044 -157.4585

Random effects:
 Formula: ~1 | Replicate
        (Intercept) Residual
StdDev:   0.1595581 2.400773

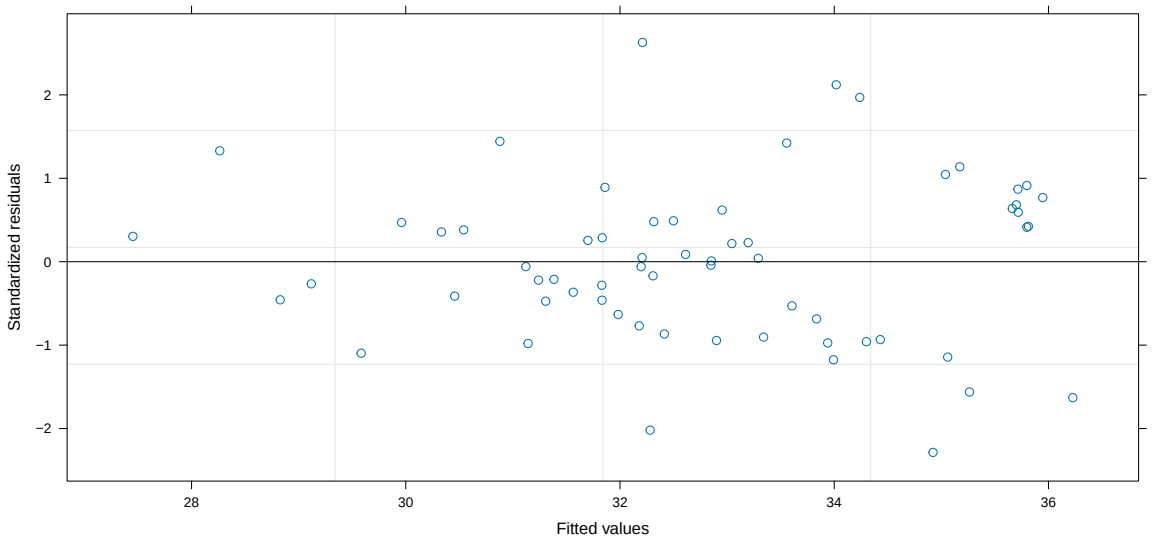
Correlation Structure: Compound symmetry
 Formula: ~1 | Replicate
 Parameter estimate(s):
        Rho
-0.02398201
Fixed effects:  C180 ~ Pressure + Temperature + Flow + Time + FA
                Value Std.Error DF   t-value p-value
(Intercept) 13.217661  5.800988 55  2.278519  0.0266
Pressure     0.003095  0.002519 55  1.228503  0.2245
Temperature  0.007544  0.014844 55  0.508243  0.6133
Flow         0.126621  0.152955 55  0.827829  0.4113
Time        -0.022738  0.010003 55 -2.273232  0.0269
FA           0.196713  0.038637 55  5.091332  0.0000
 Correlation:
            (Intr) Pressr Tmprtr Flow   Time
Pressure    -0.137
Temperature -0.806  0.008
Flow        -0.058  0.022  0.023
Time        -0.202 -0.015 -0.015 -0.044
FA          -0.446 -0.087 -0.089 -0.256  0.170
```

```
Standardized Within-Group Residuals:
         Min           Q1          Med           Q3          Max
-2.285822216 -0.659311273  0.009398242  0.541413528  2.628278708

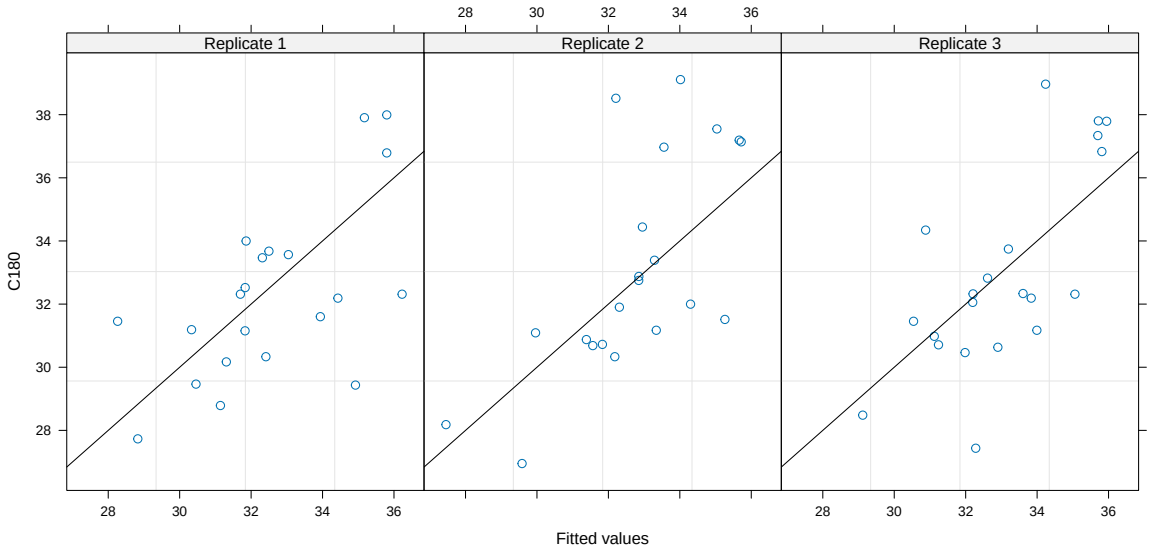
Number of Observations: 63
Number of Groups: 3
```

Various diagnostic plots for `mdl2`:

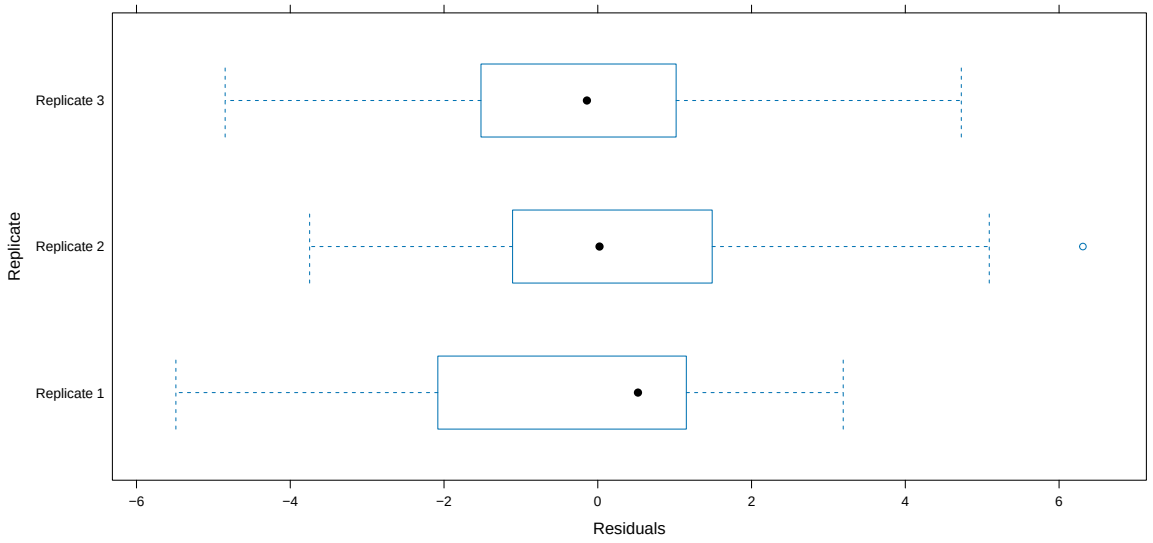
```
# residuals vs fitted (default), should be uniformly scattered
# around the zero line
plot(mdl2)
```




```
# observed vs fitted, should be distributed around the identity line
plot(mdl2, C180 ~ fitted(.) | Replicate, abline = c(0,1))
```




```
# boxplot of residuals by replicate, should all be centred around zero
# and with similar variance
plot(mdl2, Replicate ~ resid(.))
```



Because pressure, temperature and flow were seen to be not associated with % C 18:0 in model 2, we refit model 2a only with time and FA as fixed effects.

```r
mdl2a <- nlme::lme(fixed = C180 ~ Time + FA,
            random = ~ 1 | Replicate,
            data = df,
            method = "REML", na.action = na.omit,
            correlation = corCompSymm(form = ~ 1 | Replicate))
summary(mdl2a)
```

```
Linear mixed-effects model fit by REML
  Data: df
       AIC      BIC    logLik
  310.6635 323.2296 -149.3318

Random effects:
 Formula: ~1 | Replicate
        (Intercept) Residual
StdDev:   0.1610843 2.388083

Correlation Structure: Compound symmetry
 Formula: ~1 | Replicate
 Parameter estimate(s):
        Rho
-0.02333568
Fixed effects:  C180 ~ Time + FA
                Value Std.Error DF   t-value p-value
(Intercept) 16.683741  3.328317 58  5.012666  0.0000
Time        -0.022139  0.009935 58 -2.228375  0.0297
FA           0.210298  0.036881 58  5.701989  0.0000
 Correlation:
     (Intr) Time
Time -0.377
FA   -0.973  0.164

Standardized Within-Group Residuals:
       Min         Q1        Med         Q3        Max
-1.9933070 -0.5453859 -0.0936394  0.6250807  2.5533851

Number of Observations: 63
Number of Groups: 3
```

The comparison between model 2 and model 2a in terms of information criteria (AIC, BIC) suggests that model 2 with four fixed effects (pressure, temperature, flow, time) is able to

explain more variability in the outcome, despite three fixed effects not being significantly associated with the outcome.

Additionally, model 1 suggests that flow is significantly associated with the total fraction of fatty acids (FA). Based on this, it may not be appropriate to include FA as covariate in model 2 in addition to flow, because it would be equivalent to regress flow out twice. Therefore we refit model 2 without FA:

```
mdl2b <- nlme::lme(fixed = C180 ~ Pressure + Temperature + Flow + Time,
            random = ~ 1 | Replicate,
            data = df,
            method = "REML", na.action = na.omit,
            correlation = corCompSymm(form = ~ 1 | Replicate))
summary(mdl2b)
```

```
Linear mixed-effects model fit by REML
  Data: df
       AIC      BIC    logLik
  347.9323 364.4159 -165.9662

Random effects:
 Formula: ~1 | Replicate
        (Intercept) Residual
StdDev:   0.1698481 2.873519

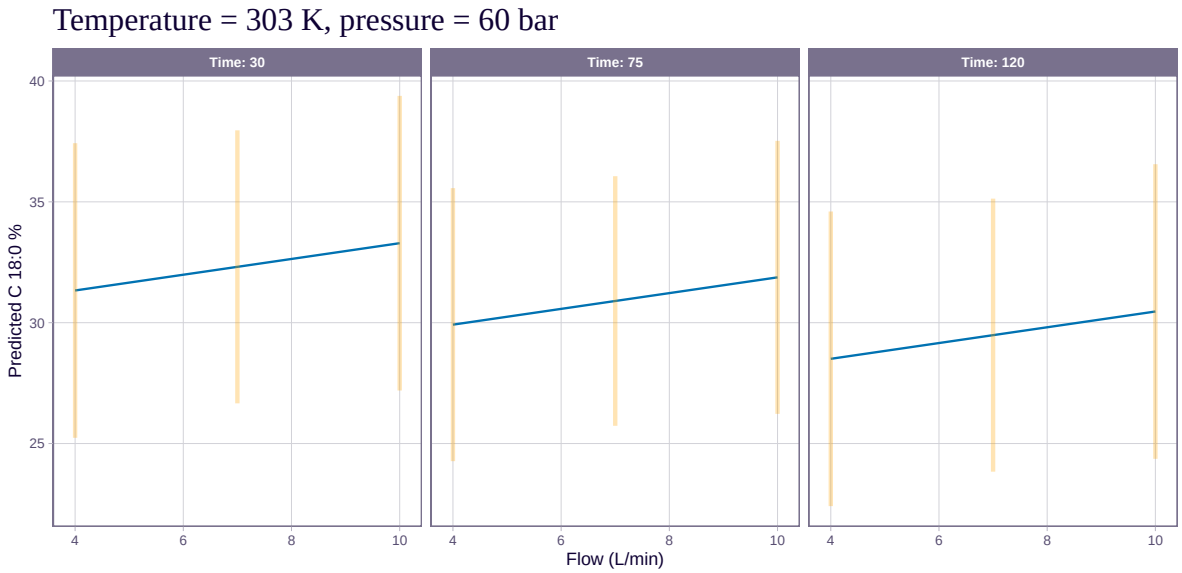
Correlation Structure: Compound symmetry
 Formula: ~1 | Replicate
 Parameter estimate(s):
        Rho
-0.02919287
Fixed effects:  C180 ~ Pressure + Temperature + Flow + Time
                Value Std.Error DF   t-value p-value
(Intercept) 26.391094  6.228820 56  4.236933  0.0001
Pressure     0.004212  0.003011 56  1.398701  0.1674
Temperature  0.014276  0.017741 56  0.804697  0.4244
Flow         0.326175  0.177411 56  1.838529  0.0713
Time        -0.031408  0.011827 56 -2.655534  0.0103
 Correlation:
            (Intr) Pressr Tmprtr Flow
Pressure    -0.197
Temperature -0.948  0.000
Flow        -0.199  0.000  0.000
Time        -0.142  0.000  0.000  0.000
```

```
Standardized Within-Group Residuals:
        Min          Q1         Med          Q3         Max
-2.30368407 -0.59399033 -0.02457946  0.55197474  2.35044678

Number of Observations: 63
Number of Groups: 3
```

In this model 2b both time and flow have a strong effect on the % C 18:0. Additionally, the comparison between model 2 and model 2b in terms of information criteria (AIC, BIC) confirms that model 2b provides a better fit than model 2.

Estimated marginal means of percentage of C 18:0 in the extracted sample, at fixed temperature of 303 K and pressure of 60 bar, for flow rates of 4, 7, and 10 L/min, and times of 30, 75, and 120 minutes:



Numerical values of the estimated marginal means (i.e., predictions):

Flow	Time	yvar	SE	df	LCL	UCL	tvar	xvar
4	30	31.3	1.42	2	25.2	37.4	1	4
4	75	29.9	1.31	2	24.3	35.6	1	4
4	120	28.5	1.42	2	22.4	34.6	1	4
7	30	32.3	1.31	2	26.7	38.0	1	7
7	75	30.9	1.20	2	25.7	36.1	1	7
7	120	29.5	1.31	2	23.8	35.1	1	7
10	30	33.3	1.42	2	27.2	39.4	1	10

```
   10   75 31.9 1.31  2 26.2 37.5 1      10
   10  120 30.5 1.42  2 24.4 36.6 1      10

Degrees-of-freedom method: containment
Confidence level used: 0.95
```

```
sessionInfo()
```

```
R version 4.4.3 (2025-02-28 ucrt)
Platform: x86_64-w64-mingw32/x64
Running under: Windows 11 x64 (build 26100)

Matrix products: default


locale:
[1] LC_COLLATE=English_United Kingdom.utf8
[2] LC_CTYPE=English_United Kingdom.utf8
[3] LC_MONETARY=English_United Kingdom.utf8
[4] LC_NUMERIC=C
[5] LC_TIME=English_United Kingdom.utf8

time zone: Europe/London
tzcode source: internal

attached base packages:
[1] grid      stats     graphics  grDevices utils     datasets  methods
[8] base

other attached packages:
 [1] emmeans_2.0.0    nlme_3.1-167     ggcorrplot_0.3.0 GGally_2.4.0
 [5] kableExtra_1.4.1 knitr_1.50       lubridate_1.9.4  forcats_1.0.0
 [9] stringr_1.6.0    dplyr_1.2.0      purrr_1.2.1      readr_2.1.5
[13] tidyr_1.3.1      tibble_3.3.1     ggplot2_4.0.3    tidyverse_2.0.0
[17] readxl_1.4.5

loaded via a namespace (and not attached):
 [1] gtable_0.3.6       xfun_0.51          lattice_0.22-6     tzdb_0.5.0
 [5] vctrs_0.7.1        tools_4.4.3        generics_0.1.4     sandwich_3.1-1
 [9] pkgconfig_2.0.3    Matrix_1.7-4       RColorBrewer_1.1-3 S7_0.2.2
```

```
[13] lifecycle_1.0.5    compiler_4.4.3     farver_2.1.2       textshaping_1.0.1
[17] codetools_0.2-20   htmltools_0.5.8.1  yaml_2.3.10        pillar_1.10.2
[21] MASS_7.3-64        viridis_0.6.5      multcomp_1.4-28    ggstats_0.14.0
[25] tidyselect_1.2.1   digest_0.6.37      mvtnorm_1.3-3      stringi_1.8.7
[29] reshape2_1.4.4     labeling_0.4.3     splines_4.4.3      fastmap_1.2.0
[33] cli_3.6.5          magrittr_2.0.3     survival_3.8-3     TH.data_1.1-3
[37] withr_3.0.2        scales_1.4.0       timechange_0.3.0   estimability_1.5.1
[41] rmarkdown_2.29     gridExtra_2.3      cellranger_1.1.0   zoo_1.8-14
[45] hms_1.1.3          coda_0.19-4.1      evaluate_1.0.3     viridisLite_0.4.2
[49] rlang_1.1.7        Rcpp_1.0.14        xtable_1.8-4       glue_1.8.0
[53] xml2_1.3.8         svglite_2.2.2      rstudioapi_0.17.1  jsonlite_2.0.0
[57] plyr_1.8.9         R6_2.6.1           systemfonts_1.3.2
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